Introduction to STL

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week 5

課程簡報

Outline

Recap Problems

Introduction

What is STL?

STL stands for Standard Template Library, which is a powerful set of C++ template classes and functions that provide data structures and algorithms. It provides a collection of tools for handling any type of data.

The most poworful feature of the STL is that it allows you handle any type of data, e.g., int, double, string, and even another STL container. This makes it a very versatile and powerful tool for C++ programming.

Lecture arrangement

This is a series of lectures that will introduce you to the STL and how to use it effectively in your C++ programming.

In these lectures, we'll cover some common STL components, but not ALL OF THEM. The STL is vast, and we will focus on the most commonly used parts that are essential for competitive programming and general C++ development.

Stack

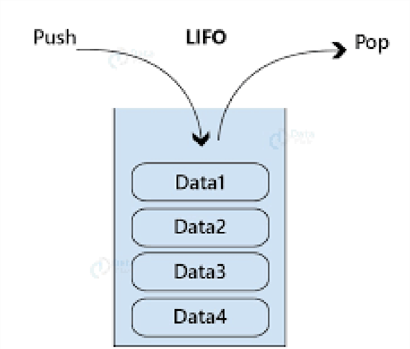
What is stack?

In English, stack means ”a pile of things arranged one on top of another(Cambridge Dictionary)”. In C++, stack is a container adapter that gives the programmer the functionality of a stack data structure.

How a stack works?

Stack is a Last In First Out (LIFO) data structure. The last element added to the stack will be the first one to be removed.

In other words, you can only access the top element of the stack(i.e., the most recently added element). You can push elements onto the stack, remove elements from the stack, and check the top element of the stack.



Here are some common operations of stack

```
stack<int> s; //create a stack of integers
s.push(1); //push 1 onto the stack
s.push(2); //push 2 onto the stack
cout<<s.top(); //print the top element of the stack
s.pop(); //remove the top element of the stack
```

Try these codes

```
stack<string> s; //create a stack of strings
s.push("Hello"); //push "Hello" onto the stack
s.push("World"); //push "World" onto the stack
cout<<s.top(); //print the top element of the stack
s.pop(); //remove the top element of the stack
cout<<s.top(); //print the top element of the stack
```

What will be the output of the above code? Why? Is it reasonable?

Practice Problems

Practice

In this section, the following problems are provided:

- i213, a565 are simple problems
- c123, c700, j607 are complex problems
- r395 is a hard problem

Choose suitable problems depending on your level.